

Spring 2026. Physics 510.

Quantum Theory II.

Homework # 3.

Due to February 23rd, 2026.

1. **Heisenberg's principle for orbital momentum components.** Given the state $|l, m\rangle$, compute the expectation values of $\langle L_x^2 \rangle$, $\langle L_y^2 \rangle$, $\langle L_z^2 \rangle$ as well as the uncertainties ΔL_x , ΔL_y , ΔL_z and eventually $\Delta L_x \Delta L_y$, $\Delta L_x \Delta L_z$, $\Delta L_y \Delta L_z$.
2. **Gravitino. Spin-3/2 particles are absolutely necessary in a supergravity theory as superpartners of gravitons .** Find the representation matrices for $j = 3/2$.
3. Consider a total angular momentum $\vec{J} = \vec{L} + \vec{S}$, where $\vec{L}$ is an orbital angular momentum and $\vec{S}$ is a spin. Prove that $[\vec{J}, \vec{L} \cdot \vec{S}] = 0$.
4. Find a unitary transformation that diagonalize σ_x . What are new expressions for the other two Pauli matrices? Explain the results.
5. **Schwinger's representation of an angular momentum algebra. Sakurai, Napolitano. Sec. 3.9, p. 218. Desai, chapter 11, problem 2.** Do all calculations and prove the Wigner's formula Sakurai 3.426.
6. **Equivalence of two different sets of J-matrices.** We got two sets of 3 dimensional matrices of $j = 1$ generators of rotation. The first one from direct calculations of matrix elements in the $|j, m\rangle$ basis. Another one in 502 classes from the rotational matrices. Prove their equivalence, i.e. the fact that these two sets are connected by some unitary transformation. Hint: find a transformation that diagonalize J_z of the latter basis.